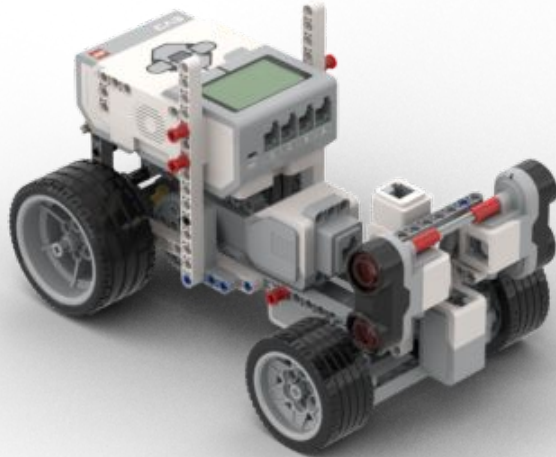
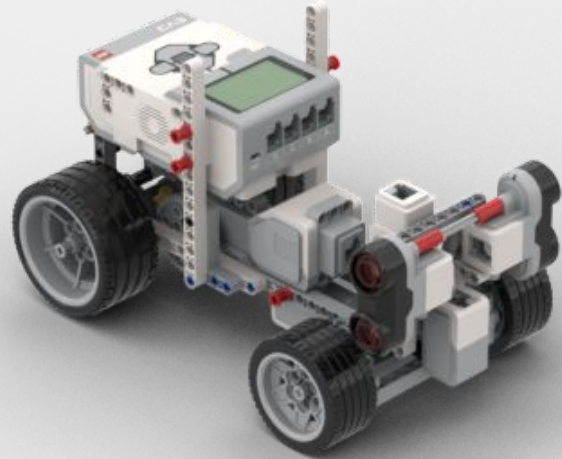


# **WRO FUTURE ENGINEERS OPEN CHALLENGE**

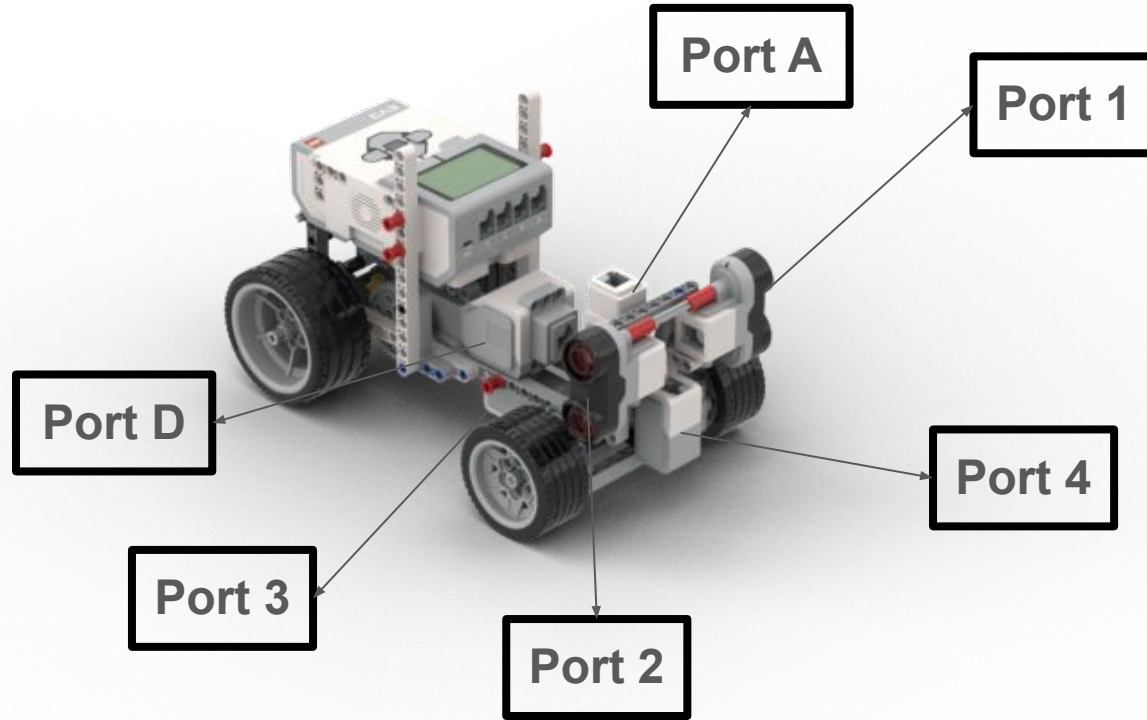


# Robot Build

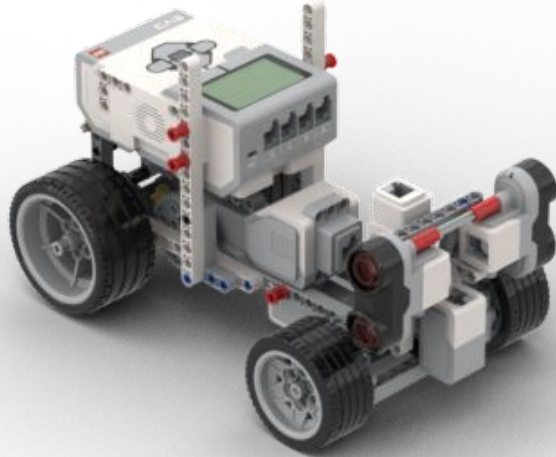
Open the “WRO Future Engineers Open Challenge Building Instruction” pdf file



# Ports

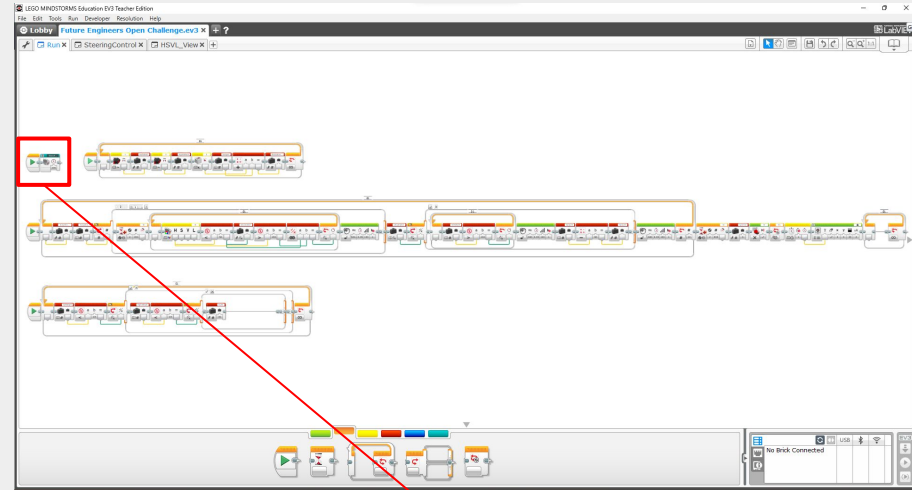


# WRO FUTURE ENGINEERS CODE EXPLAINED



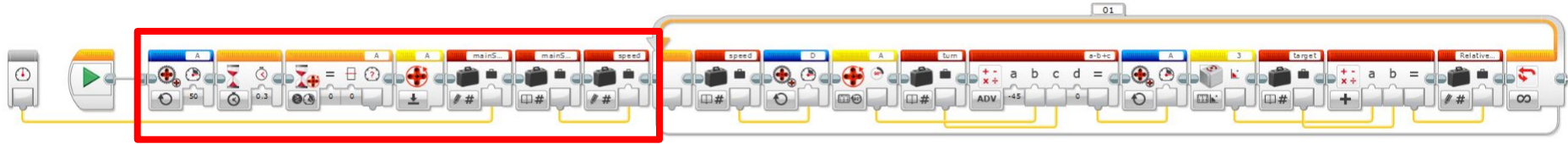
# Steering Block

This block control the steering of the robot, it runs as a multitask



# Steering Block

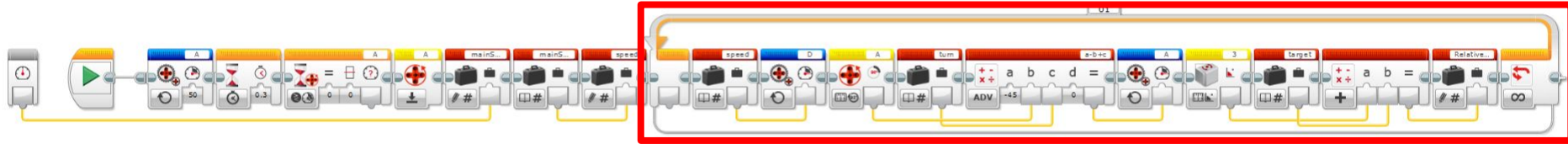
First, the robot reset the steering motor



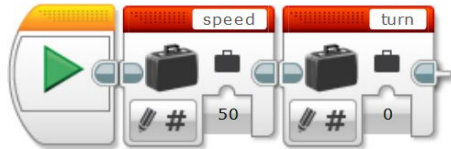
# Steering Block

The loop control the driving speed, and the turning direction.

## Change the speed and turning direction by setting the values using the variables

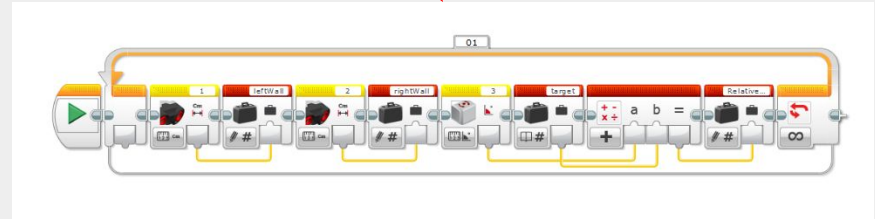
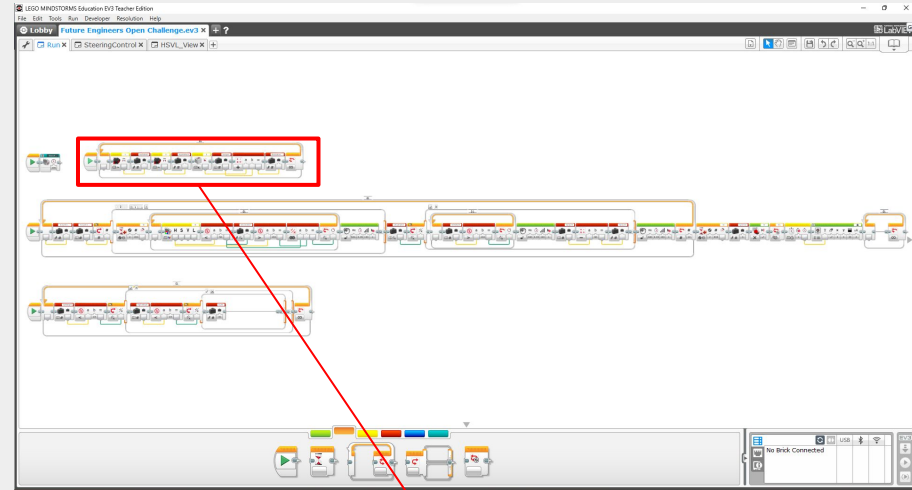


## Example:



# Sensor Reading

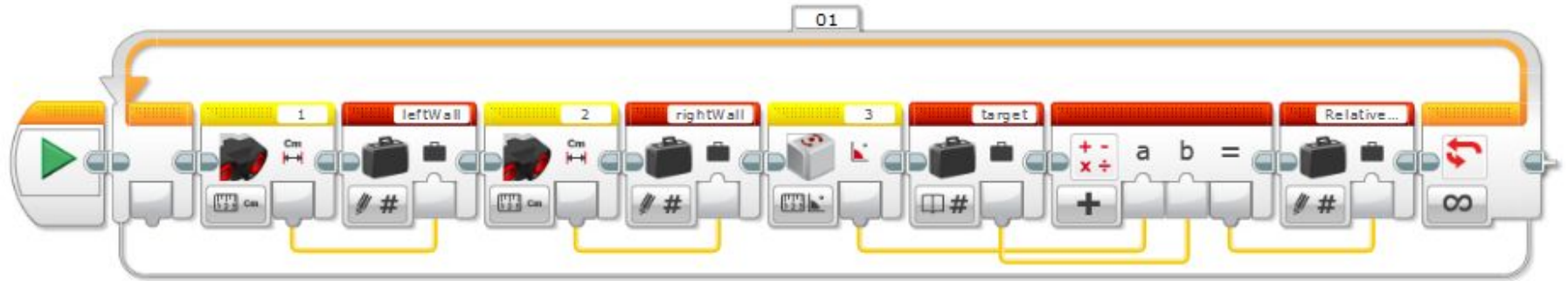
This block return all sensor value using variables





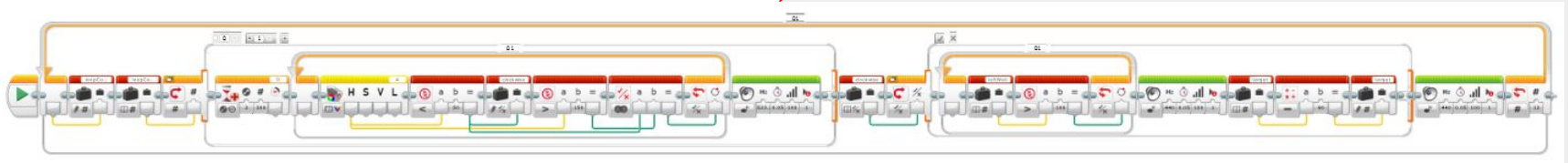
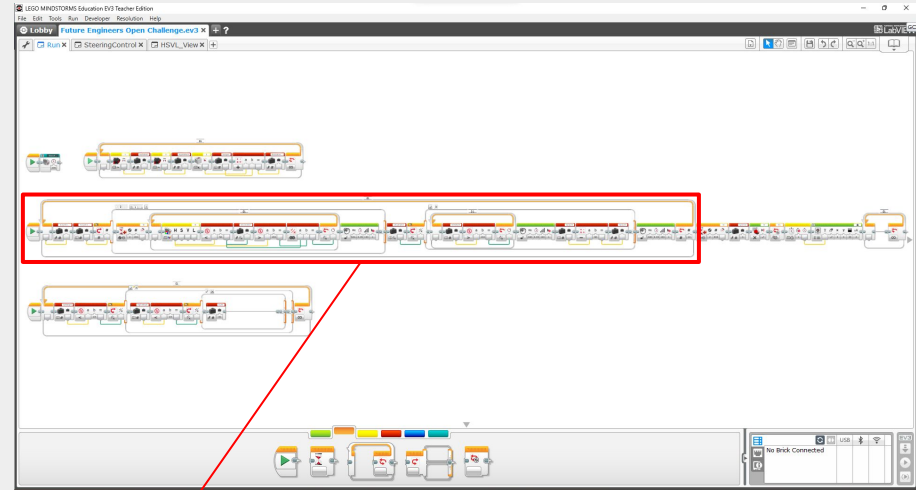
## Reading Sensor Value

This loop reads the sensor values write it into variables, these variables can be access anytime and anywhere with minimum delay



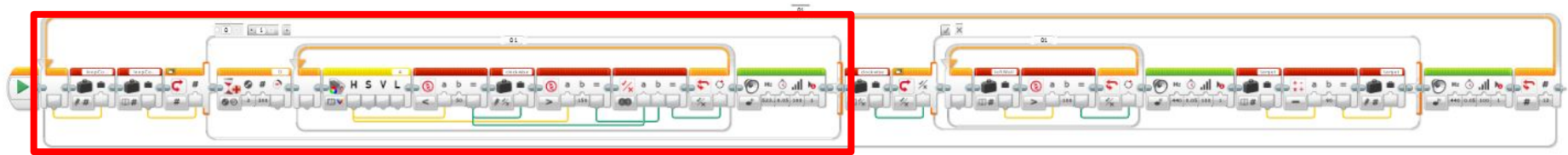
# Main Loop

The main loop controls the program for the robot to go 3 round



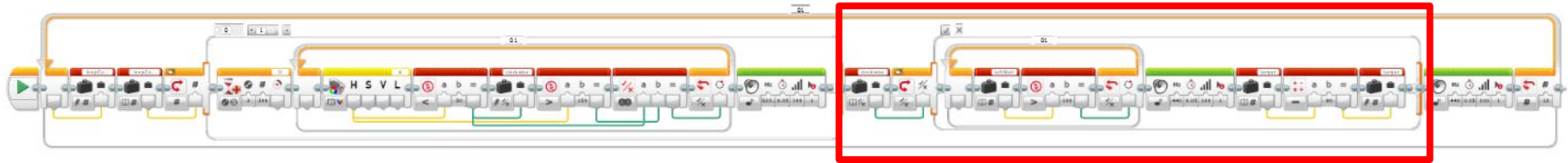
# Main Loop

During the first loop, the robot uses the color sensor to detect the floor color to determine the direction of the robot, clockwise or counter clockwise.



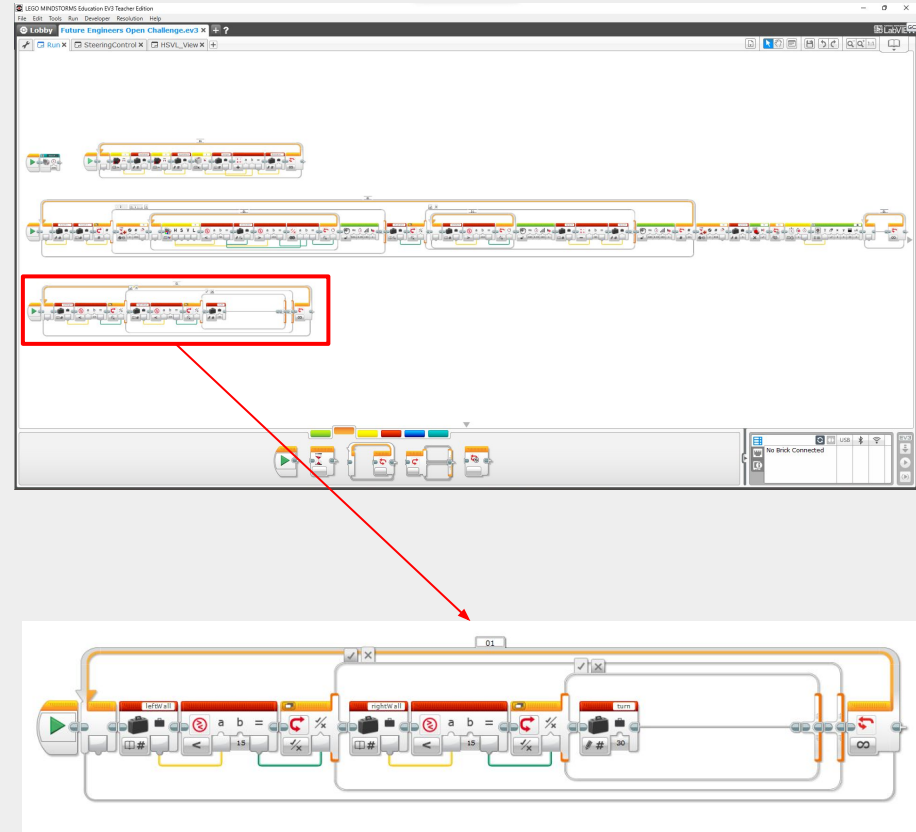
# Main Loop

Robot will run until wall is not present and change the heading direction according to clockwise or counter clockwise



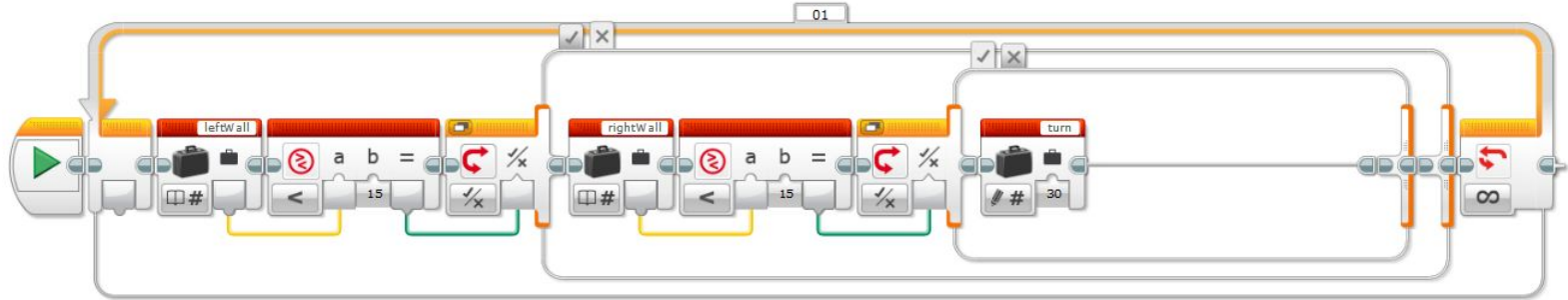
# Control Loop

The control loop control the robot heading direction



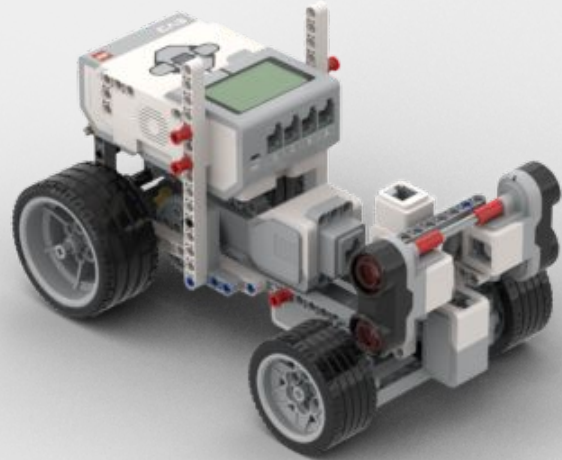
# Control Loop

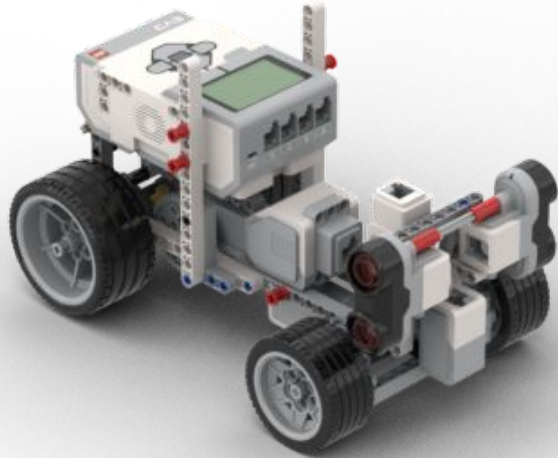
The control loop check the left and right wall to do adjustment.



# Program Run

Download and run the program,  
robot will finish 3 round and stop  
back at the starting section





**Thank You!**